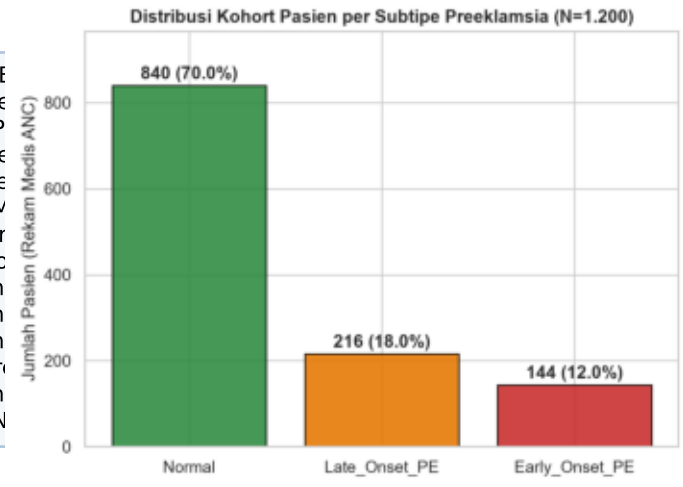
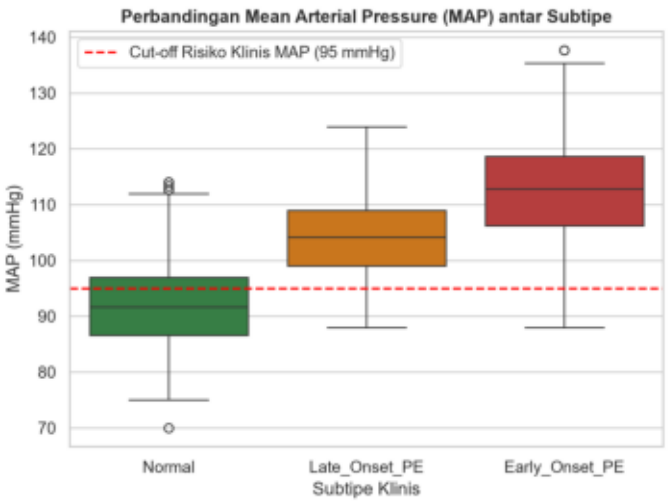
Nayyara Rauf Ghani Al Khayri (NIM: A11.2024.16055)
Program Studi Teknik Informatika, Fakultas Ilmu Komputer — UTS Data Mining 2026

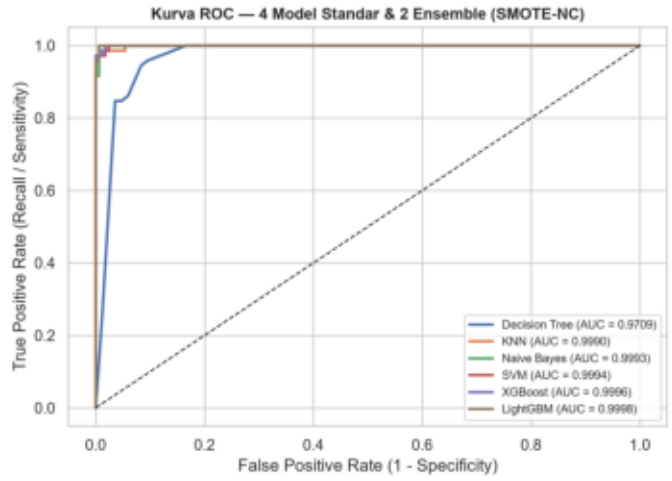


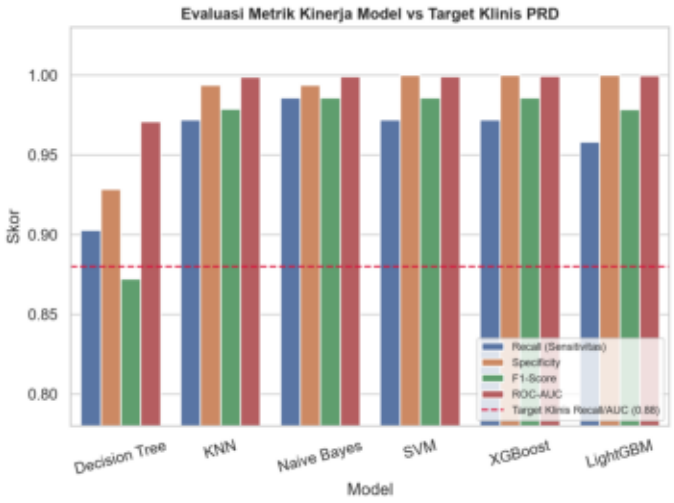


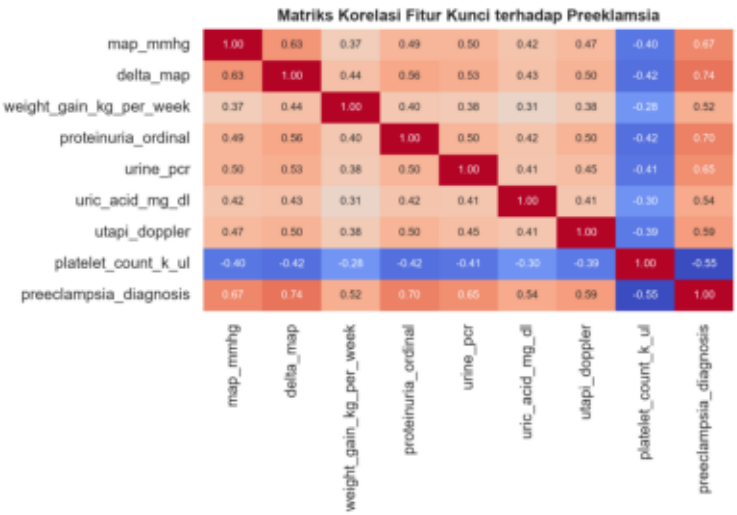
3. PERANCANGAN DATASET & PRA-PEMROSESAN DATA (KRITERIA 3)
Dataset `preeclampsia_clinical_dataset.csv` mencakup N = 1.200 pasien unik (840 Normal [70%], 216 Late-Onset PE [18%],
1) Deduplikasi 12 rekaman ganda berdasarkan `patient_id`;
2) Median Imputation pada atribut numerik yang memiliki missing values (`lila_cm`, `urine_pcr`, `uric_acid_mg_dl`, `utapi_d
3) Validasi konsistensi fisiologis tekanan darah dan kalkulasi ulang MAP = (SBP + 2*DBP)/3;
4) Ordinal Encoding pada `proteinuria_dipstick` (0 s.d. 3) dan StandardScaler di dalam Pipeline bebas data leakage.

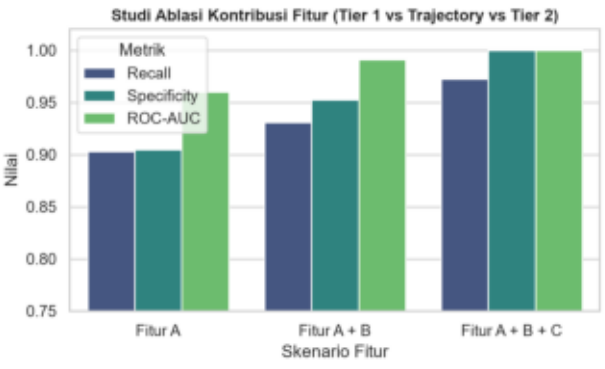
4. HASIL EKSPERIMEN KOMPARASI 4 MODEL & STUDI ABLASI FITUR

Algoritma	CV AUC	Accuracy	Recall (Sens.)	Specificity	F1-Score	Test AUC
Decision Tree	0.9536	92.08%	90.28%	92.86%	87.25%	0.9709
KNN	0.9996	98.75%	97.22%	99.40%	97.90%	0.9990
Naive Bayes	0.9994	99.17%	98.61%	99.40%	98.61%	0.9993
SVM	0.9998	99.17%	97.22%	100.00%	98.59%	0.9994
XGBoost	0.9991	99.17%	97.22%	100.00%	98.59%	0.9996
LightGBM	0.9994	98.75%	95.83%	100.00%	97.87%	0.9998









5. KESIMPULAN: Integrasi Fitur A (Tier 1), Fitur B (Trajectory), dan Fitur C (Tier 2 Biomarker) meningkatkan sensitivita